

Morphogenesis I: Coast, Shrub, Forest

Hye Min Cho
Independent
United States of America
hyeminchocho@gmail.com

Maru Garcia
Independent
United States of America
studio@marugfierro.com

Maura Palacios Mejia
Mount San Antonio College
United States of America
mpalaciosmejia@mtsac.edu

ACM Reference Format:

Hye Min Cho, Maru Garcia, and Maura Palacios Mejia. 2022. Morphogenesis I: Coast, Shrub, Forest. In *Proceedings of the ACM SIGGRAPH '22 Art Gallery, August 07-11, 2022, Vancouver, BC, Canada*. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532837.3534949>

1 PROJECT URL

<https://www.biomedistudio.com/>

2 DESCRIPTION OF ARTWORK

2.1 Introduction

Soil is the foundation for healthy ecosystems and plays an important role in sustaining life. Soil serves as the baseline for life and provides a variety of ecosystems and societal functions. Despite sustaining the world's demands of food, fibers, and fuels, regulating Earth's air and water quality, and storing greenhouse gases, the value of soil, commonly termed "dirt", is often dismissed. This human disconnect from this important source of life, especially in the middle of our climate crisis, needs to be addressed as a vital sustainability practice. Soil health is largely dependent on a diverse thriving microorganism composition. This unseen world of microorganism biodiversity (diversity of life) can be detected, enumerated, and characterized through advancement in molecular methods such as the use of Environmental DNA (eDNA). This tool takes advantage of DNA found in the environment, from organisms in the form of feces, skin cells, pollen, whole cells (microbes) that can then be collected from soil, sediment, water, and even air! We can use the eDNA from the sample to obtain information on the organisms found in soil from microbes, plants, and vertebrates. From the Greek morphê - shape, and genesis - creation, literally means "the generation of form." Morphogenesis is the process that controls the spatial organization and distribution of cells during the development of an organism. An example of this mechanism includes the transformation of a caterpillar to a butterfly, which demonstrates the ability of an organism's cells to self-organize a new body shape. However, morphogenesis also represents coordinated behavior to build and develop the shape, position, and interconnection of a structure, for example, the replicated pattern of termite mounds. Morphogenesis is a transcending concept that captivates the human imagination. Morphogenesis captures the essence of this project, by expressing how soil can be considered a complex living organism

in combination with the computational process of neural cellular automata.

2.2 CALeDNA microbial dataset & analyses

The CALeDNA program pairs volunteer community scientists with University of California researchers to collect soil, sediment, and water samples from ecosystems across California for environmental DNA (eDNA) metabarcoding analysis to assess the biodiversity of microbes, fungi, plants, and animals. Our artwork uses the CALeDNA microbial dataset was collected by community scientists across the state of California between March and June 2017 at three distinct ecosystems, coast, shrub and forest¹. The 278 collected samples were processed for three molecular markers for the amplification of microbial DNA; 16S which targets bacteria, 18S which targets eukaryotes, and FITS which targets fungi. After obtaining the sequences, the raw data was analyzed including taxonomy assignment, estimates of each group at the phylum level, and community composition by ecosystem at the phylum level. In addition, we chose representative images for each phylum to serve as the inspiration of the textures and colors for the visualization step of this project.

2.3 Computational Process for visualizing the dataset

When choosing the computational process to translate the data into the visual, one of the most important criteria for us was being able to express the beauty of soil diversity visually through form and composition. We were looking for a more organic and less digital quality, as well as a conceptual parallel between the computational process that generates the visual and the biological processes of the microorganisms in the soil.

Cellular Automata can be used to generate complex forms by applying a set of hand-crafted rules repeatedly on a grid of cells. The state of each cell at the next time step is determined by the current state of the cells in its close neighborhood. Cellular Automata, like the process of Morphogenesis, is an example of self-organization. We thought it serves as a good parallel to the biological mechanisms that occur in the soil among the microorganisms.

For our project, we chose Neural Cellular Automata (NCA) by Mordvintsev et al.² as the main process of visualizing the data. With Neural Cellular Automata we are able to train the neural network to learn a set of rules to be applied onto the cells to generate a particular type of texture or image. To account for the 20 phyla in the dataset, we aimed to modify the NCA model to generate 20 different textures or images.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '22 Art Gallery, August 07-11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9372-0/22/08.

<https://doi.org/10.1145/3532837.3534949>

¹Lin, Meixi et al. "Landscape analyses using eDNA metabarcoding and Earth observation predict community biodiversity in California." *Ecological applications: a publication of the Ecological Society of America* vol. 31,6 (2021): e02379

²Mordvintsev, et al., "Growing Neural Cellular Automata", Distill, 2020.



We duplicated the second dense layer of the NCA model 20 times, one for each phylum and its corresponding image. At each training step, a phylum category would be randomly chosen and its corresponding second dense layer along with the shared first dense layer would be optimized for the target image. The same first dense layer was always used for all categories. By sharing the first dense layer across the categories, we hypothesized the model may develop a shared internal representation which can be used by each of the second layers to generate 20 different target images. In practice, the modified model was not capable of recreating the 20 target images accurately. However, it was able to generate 20 distinct visually intriguing textures. Furthermore, when a combination of rules from different second layer outputs were applied onto the cells, the resulting visuals had a mixture of formal characteristics from all their phylum categories.

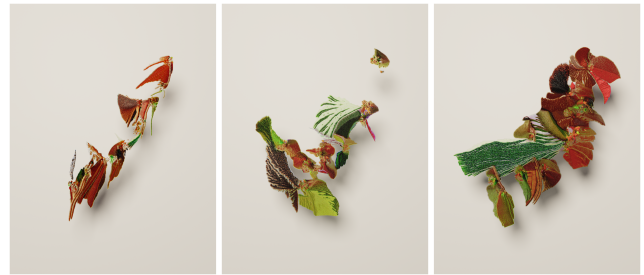


To generate the final artwork, we translated the data for each of the ecosystems into a distribution of ‘seeds’ each belonging to one of the 20 phyla. And at each time step, the cells were updated according to the rules of their categories. As the forms grow larger from the seeds and they start to overlap, a combination of rules were applied to generate forms that have a mixture of characteristics from their constituent categories. The interaction and propagation of phylum categories and their rules generate the distinct final forms for each of the three ecosystems.

2.4 Conclusion

This project integrates science, art, and technology to visually demonstrate the importance of soil as a source of life through the dynamic microbial biodiversity in different California ecosystems. Each visual made to generate the final art piece for each ecosystem represents the network of interactions occurring at the microbial level, which is normally unseen. In unity, the final artwork represents the accumulation of unique microbial diversity at the individual sites as well as the ecosystem scale towards maintaining healthy soil. As we continue to better understand the importance of soil, we can become better stewards of this living element in our aquatic and terrestrial landscapes.

Morphogenesis I: coast, shrub, forest is a commissioned piece by the California Natural Resources



3 ARTIST INFORMATION

Hye Min Cho (b. Incheon, South Korea) is a media artist, who uses custom software to visualize the experiences at odds with the normal and the legitimate. Her works are renditions of reality transcoded through computation aiming to visualize the information that may not have practical significance to the human eyes. Hye Min holds a M.F.A in Design | Media Arts from UCLA and a B.Sc in Electrical Engineering and Computer Science from UC Berkeley. <https://hyeminchocho.com/>

Maru García is a transdisciplinary artist and researcher working across art + science + environment. Her work highlights the importance of eco-aesthetics, where relationships and community are proposed to build cultures of regeneration. Maru holds an MFA in Design & Media Arts from UCLA as well as an M.S. in Biotechnology and a B.S. in Chemistry both from Tecnológico de Monterrey, México. Currently, she is based in LA. <https://marugfierro.com/>

Dr. Maura Palacios Mejia is an Assistant Professor of Biology at Mount San Antonio College. Her research focuses on applying environmental DNA (eDNA; DNA found in the environment) to address issues in conservation and restoration. Dr. Palacios Mejia obtained her doctorate degree in Wildlife & Fisheries Sciences from Texas A&M University, M.S. degree in Biology from California State University, Los Angeles and B.S. in Marine Biology & Zoology from California State University, Long Beach.